Encouraging the Mutual Interact Between Dataset-Level and Image-Level Context for Semantic Segmentation of Remote Sensing Image

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***Abstract*— Recently, semantic segmentation of remote sensing images has witnessed rapid advancement with the adoption of deep neural networks. Contextual cues, referring to the long-range correlation between pixels, are crucial for achieving accurate segmentation results, particularly for objects with less discriminative characteristics in these images. Currently, most studies are centered on incorporating contextual cues by aggregating context information at the dataset level or image level. However, current research often treats contextual cue modeling at the dataset-level and image level as independent procedures, neglecting the intrinsic correlation between these two feature levels. Consequently, the obtained contextual cues are suboptimal. This issue is particularly critical in the semantic segmentation of remote sensing images. To address this, we propose to encourage mutual interaction between dataset-level and image-level contextual cues. Firstly, we propose an interactive dataset–image context aggregation scheme to obtain complementary and consistent multilevel contextual cues. Additionally, we introduce a parallel feature interaction network (PFI-Net) that progressively extracts and fuses features across multiple layers, enabling effective integration of multilevel contexts. Furthermore, we introduce an enhanced shifted window-based cross-attention mechanism to augment model efficiency. Extensive experimental results on the widely used Vaihingen dataset, GaoFen-2 dataset, and instance segmentation in aerial images dataset (iSAID) effectively demonstrate the superiority of our proposed method over the other state-of-the-art methods.**

***Index Terms*— Contextual cue, remote sensing image, semantic segmentation.**

I. Introduction

HIGH-RESOLUTION remote sensing (HRS) semantic segmentation aims to achieve pixel-level classification and discrimination in HRS images, representing a significant research direction in the field of remote sensing applications [1], [2], [3]. Serving as the foundation for extracting valuable information from remote sensing data, this approach plays a pivotal role in object-oriented image analysis [4]. The quality of segmentation significantly influences the accuracy and effectiveness of subsequent information extraction processes [5]. In practical terms, there exists a critical necessity to accurately and promptly obtain feature information from remote sensing images. This requirement is applicable in various domains, including ecological monitoring, land use analysis, and geographic mapping. Therefore, the achievement of refined pixel-level semantic segmentation in HRS images is of immense significance [6].

Over the past decades, semantic segmentation of HRS images utilizing convolutional neural networks (CNNs) [7], [8], [9], [10] has produced fruitful research results and has emerged as a mainstream method in remote sensing image segmentation [11], [12], [13], [14], [15]. Furthermore, deep learning frameworks in natural language processing have taken a different path, where the prevalent architecture today is instead the transformer [16], [17]. Unlike CNNs, the transformer is adept at modeling global long-distance dependencies, unconstrained by the limitations of receptive field. The transformer's remarkable success in natural language processing has significantly influenced the field of image analysis. Several studies have already adapted the transformer to image segmentation, yielding promising results [18], [19].

Contextual cues typically refer to the long-range correlation between pixels [20], [21]. For example, in remote sensing scenes, a boat can often be mistaken for a car due to the similar appearance of objects. However, when considering the contextual cue, such as a scene being described as a "boathouse" near a river, a more accurate prediction can be yielded. Early studies of contextual cues focused mainly on modeling multiscale cues within a single image, termed image-level context [22]. Representative works, including Deeplab [23], [24], [25], OCRNet [21], Segmenter [26], and ISNet [27], have been widely employed due to their solid performance. Though impressive, these image-level context schemes focus solely on aggregating contextual information from a single image's perspective, ignoring representations of the same category across multiple images. This issue becomes particularly evident in remote sensing images, characterized by significant intra-class variation and inter-class similarity. For example, the pixels in the regions of objects of small

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scales tend to capture much more contextual information from the regions of other objects. Too much contextual information from other objects may cause the network to mislabel these pixels as other categories. Recently, works such as MCIBI [28] and MCIBI++ [22] have explored dataset-level context beyond a single image, aggregating contextual cues with higher weights. Nevertheless, some studies [28], [29], [30] suggest that the globally biased dataset-level context lacks detailed discrimination power compared to image-level context.

We propose to investigate the context aggregation scheme along the line of exploring the relation between dataset-level and image-level context for complementarily augmenting the contextual representation. Our motivation for this work stems from our finding of two significant shortcomings in existing methods. On the one hand, existing methods treat contextual cue modeling at the dataset level and image level as separate procedures. Consequently, these methods fail to achieve the goal of "complementarity." On the other hand, existing methods aggregate multilevel (dataset and image) contextual cues through simple feature concatenation along the feature channel dimension. This simplistic fusion approach leads to wastage of contextual information. More critically, it may produce conflicting information during fusion, resulting in subpar segmentation performance. In this article, we propose encouraging mutual interaction between dataset-level and image-level contexts, integrating an interactive dataset–image context aggregation scheme with a parallel feature interaction network (PFI-Net).

The interactive dataset–image context aggregation scheme aims to obtain complementary and consistent multilevel contextual cues. The main idea is illustrated in Fig. 1. The pixel-level classification capability of a deep learning-based semantic segmentation model stems from two aspects: the image-level context within a current single image and the dataset-level context encoded in a historical image set. The dataset-level context, modeled using high-dimensional information [28], aids in the modeling of the image-level context. At the same time, the image-level context not only supplements an enhanced capacity for detail discernment but also updates the dataset-level context. These interactions thus render the multilevel contexts more reliable and comprehensive.

These interactions thus enhance the overall reliability and comprehensiveness within context-centric framework.

Specifically, our aggregation scheme consists of three main steps. First, inspired by the variational autoencoder (VAE), a memory module [31], [32] is specially designed to store the distribution information from the historical image set. The stored information is dynamically updated, generating dataset-level contextual cues. Second, the original features representations are modulated and augmented with dataset-level contextual cues. Lastly, the dataset-level contextual cue is further utilized to assist the fusion of image-level context. The fusion weight factors are calculated according to the relations across different contextual levels.

The proposed PFI-Net aims to develop a suitable optimization strategy for feature fusion of multilevel contextual cues.

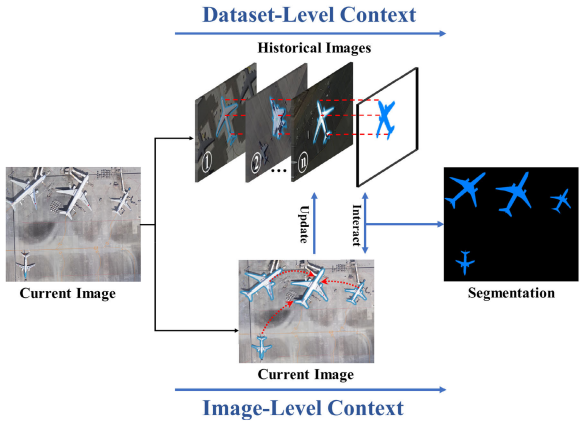


Fig. 1. Mutual interaction between dataset-level and image-level context. The upper branch denotes dataset-level context encoded from historical images and the lower branch stands for image-level context from current image. The interactions between them enable the multilevel contexts more consist and complementary.

In contrast to the "crude" approach of dimension concatenation, we have designed an interactive network comprising a stack of multiple fusion subnetworks. Parallel features derived from various contextual cues progressively enhance semantic consistency, culminating in a compatible and well-structured fusion.

A key design element of our interaction network is the cross-swin transformer (CST) block. The attention mechanism of transformer provides a potential application in fusion due to its outstanding capacity of capturing relevance in multilevel contextual cues. However, existing transformer models primarily concentrate on large-scale feature extraction, often neglecting the correlations between features from different contextual levels. Consequently, we propose the CST, which computes correlations based on improved shifted window-based cross attention. Empirical study verifies that such a design of the CST dramatically improves the fusion quality.

Moreover, the proposed model demonstrates superior efficiency compared to the conventional swin transformer [19], with the following optimization details: 1) differentiated window partition; 2) limitations on the number of transformer layers in a CST block; and 3) elimination of the patch partition layer and patch merging layer. As a result, this streamlined window-based interaction network significantly enhances the accuracy and efficiency of feature fusion.

We perform ablation studies to better understand the proposed model and validate the effectiveness of each module. In addition, the proposed method is validated on Vaihingen dataset [13] and GaoFen-2 dataset [33]. After that, it is extended to the large-scale iSAID dataset [34] for further study. Finally, the contributions of this article are summarized as follows.

1) We propose an interactive dataset–image context aggregation scheme, which effectively aggregates the complementary and consistent dataset-level and image-level contextual cues.
2) A PFI-Net is presented to progressively fuse the features across different contextual levels, yielding combined and well-structured contextual cues.

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3) We design a simple yet effective CST block based on improved shifted window-based cross attention, which dramatically boosts the overall model efficiency.
4) State-of-the-art results are achieved on three widely used HRS image semantic segmentation benchmarks, including ISPRS Vaihingen, GaoFen-2, and iSAID.

The rest of this article is organized as follows. Section II presents related work. In Section III, we introduce our proposed method. Then in Section IV, we demonstrate and analyze the method with experimental results. Finally, Section V concludes the article.

II. Related Work

A. Semantic Segmentation

Since the pioneering work of full convolutional neural networks (FCNs) [35], deep neural network-based approaches have achieved significant success in semantic segmentation. Current semantic segmentation methods primarily address the limitations of FCN, particularly the optimization of its coarse predictions. Researchers' efforts in this area can be mainly categorized into two groups. One group is focused on designing improved backbone networks to obtain more reliable features [3], [36], [37]. Addressing the challenges of gradient vanishing and model degradation in deep networks, He et al. [9] introduced ResNet, a modular architecture designed to capture cross-feature interactions and learn diverse representations by leveraging channel attention on different network branches. ResNet remains the most extensively employed backbone network to date. Furthermore, subsequent advancements, such as ResNeXt [38], ResNeSt [39], etc., have also been proposed. Additionally, there are other researches, such as ViT [18] and swin-transformer [19], that extract deep features based on transformer, incorporating excellent global properties and modal fusion capabilities [40], [41], [42], [43], [44], [45].

Another group focuses on the issue of limited receptive domain in FCN to aggregate more reliable contextual cues. Various methods have been proposed to obtain multiscale contextual cues [46], [47], [48], [49], [50]. PSPNet [51] introduces SPP, which captures hierarchical global information of different scales and subregion relationships. The Deeplab series [23], [24], [25] introduce atrous spatial pyramid pooling (ASPP) to capture multiscale context information. OCRNet [21] groups pixels into category regions based on correlation and then splices the object context feature representation (OCR) with deep network feature representations. Segmenter [26] utilizes transformer (ViT) and extends it to semantic segmentation for global context aggregation. Furthermore, ISNet [27] proposes to aggregate more valuable contextual information for each pixel from a semantic perspective to further improve pixel representation.

While these approaches have shown impressive results, they are limited to the context aggregation scheme within a single image, that is, image level. Some subsequent works, such as MCIBI [28], have recognized the importance of discriminative dataset-level context and proposed the earliest soft mining paradigm for cross-image context information, which serves as inspiration for our work. There are also recent works that share similar motivations with our work, such as MCIBI++ [22], an improved version of MCIBI. Some of these approaches focus on updating memory modules, while others extend the traditional contrast learning strategy to image semantic segmentation, aiming to learn intra-class compact and inter-class dispersed features from the perspective of the entire dataset. Our work, EMIDI, differs from these approaches in terms of motivation. Instead of "mining," our fundamental motivation is to optimize "interaction." In other words, we concentrate on the consistency of different level contextual cues and emphasize the effective fusion of features to avoid conflicting information.

B. Transformer-Based Semantic Segmentation

The pioneering work of ViT [18] introduced a transformative paradigm for computer vision processing based on transformers, achieving state-of-the-art results in semantic segmentation tasks without requiring any manual components. Subsequent research can be mainly divided into two paths. One path of investigation focuses on addressing the issues of computation complexity associated with ViT, from various perspectives to enhance training efficiency and feature extraction capabilities [19], [52], [53], [54], [55]. Its improvement mainly focuses on the design of informative category queries. The representative method is swin-transformer [19].

Another path of research concentrates on the specific downstream task of semantic segmentation, centering on the mutual integration of transformer with conventional semantic segmentation modules. For instance, methods such as SETR [56] and its improved variants employ the transformer layer for multiscale feature fusion. TransUNet [57] draws inspiration from the conventional U-Net architecture and effectively utilizes the attention mechanism's ability for long-distance context modeling. Other similar methods, including XMorpher [45], also fall into this category. On the other hand, SegFormer [58] stands out for its innovative approach in complexity reduction.

The PFI-Net proposed in our work is also essentially a transformer-based semantic segmentation approach, which provides an optimization strategy that balances efficiency and effectiveness.

III. Method

In this section, as demonstrated in Fig. 2, we first introduce the overall formulation of our framework named EMIDI. Then, we describe the details of the proposed interactive dataset–image context aggregation scheme, which incorporates the dataset-level context aggregation, dataset-level feature modulation (DLFM), and multilevel context interaction (MLCI). Finally, the internal structure of our proposed PFI-Net and CST is elaborated.

A. Overview

Given an input RGB image I, we first use a backbone $\mathcal{F}_B$ to transform I into a non-linear embedding space. Then we obtain the original pixel-level feature representations R

$$R = \text{Conv}(\mathcal{F}_B(I)) \tag{1}$$

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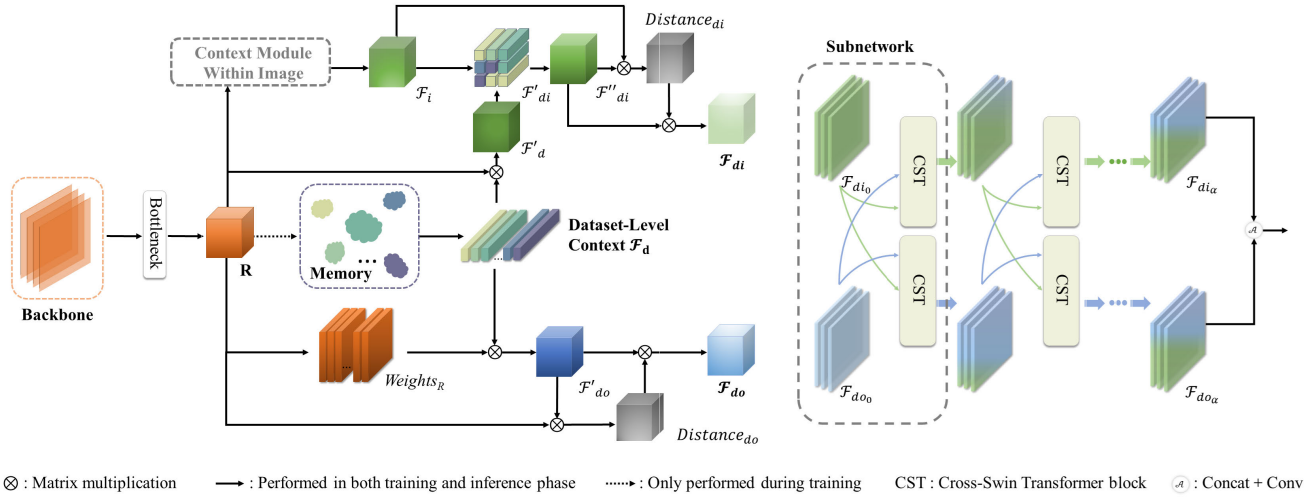


Fig. 2. Overview of the proposed framework (EMIDI). (Left) Dataset-image-level interactive context aggregation scheme, including dataset-level context aggregation, DLFM, and MLCI to accompany interaction in contextual feature extraction. (Right) PFI-Net, progressive interactive feature fusion.

where the size of R is $(H/8) \times (W/8)$, the number of channels is C, and the Conv is a convolutional layer for dimension adjustment. After that, the dataset-level context is aggregated

$$\mathcal{F}_d = \mathcal{M}_c(\mathcal{M}_d(R,\ N)) \tag{2}$$

where $\mathcal{F}_d$ is a matrix of size $N \times C$, N is the number of categories. $\mathcal{M}_d$ is the process that memory module stores and updates the distribution of semantic information in R. Then $\mathcal{M}_c$ aggregates the above distribution into dataset-level context cue $\mathcal{F}_d$. After that, the $\mathcal{F}_d$ obtained by memory module is used to interact with the original representations R, that is, DLFM $\mathcal{J}$

$$\mathcal{F}_{do} = \ \mathcal{J}(\mathcal{F}_d\ ,\ \mathcal{W}(R)) \tag{3}$$

where the matrix size of $\mathcal{F}_{do}$ is $C \times (H/8) \times (W/8)$, $\mathcal{W}$ is a simple classifier for predicting the category probability distribution weights Wights of the representations R. Meanwhile, the MLCI $\mathcal{G}$ is utilized. The purpose is to integrate the dataset-level context cue $\mathcal{F}_d$ into the aggregation process of image-level context cue $\mathcal{F}_{di}$

$$\mathcal{F}_{di} = \ \mathcal{G}(\mathcal{F}_d\ ,\ \mathcal{I}(R)) \tag{4}$$

where $\mathcal{I}$ is used to initially aggregate the image-level context of representation R. We would like to keep this step as efficient as possible. Therefore, a decoder of existing segmentation frameworks (e.g., UperNet) is utilized. It is a simple yet effective operation that both ensures that we have the optimal solution for image-level context aggregation at all times, and also empowers the network with the potential to be extended to other downstream tasks.

After the interactive dataset–image context aggregation scheme, we have obtained a set of complementary and consistent multilevel contextual cues $\mathcal{F}_{do}$ and $\mathcal{F}_{di}$. These contextual cues can already be utilized for yielding satisfactory segmentation labels. However, further development still lies ahead in terms of interaction. To address this, we develop a PFI-Net to progressively incorporate the multilevel contextual cues

$$\begin{aligned} \mathcal{F}_{do_\alpha} &= \ C_{\mathrm{ST}}\left(\mathcal{F}_{do_{\alpha-1}}, \mathcal{F}_{di_{\alpha-1}}\right) \\ \mathcal{F}_{di_\alpha} &= \ C_{\mathrm{ST}}\left(\mathcal{F}_{di_{\alpha-1}}, \mathcal{F}_{do_{\alpha-1}}\right) \end{aligned} \tag{5}$$

where α denotes the layer of the parallel subnetwork, and each subnetwork consists of two CST blocks. The details are shown in E. The contextual cues $\mathcal{F}_{do_{\alpha-1}}$ and $\mathcal{F}_{do_{\alpha-1}}$ obtain their mutual attention through two CST blocks by exchanging the order of inputs. Then the two outputs with the other's attention return to the original pipelines and prepare for next deeper interaction. There are three times of interaction in total for sufficient mutual information.

Finally, the outputs with a strong ability to represent the correspondences between contextual cues will be fused and fed into a classifier to obtain fine-grained pixel-level segmentation labels

$$O = \ \mathrm{Upsamples}\left(\mathcal{H}\left(\mathcal{A}(\mathcal{F}_{do_\alpha} \oplus \mathcal{F}_{di_\alpha})\right)\right) \tag{6}$$

where $\oplus$ denotes channel splicing, $\mathcal{A}$ is a transformation function used to adjust the channel of the input matrix to C. $\mathcal{H}$ is a classifier, O is a probability distribution of size $N \times H \times W$ storing the category of each pixel.

B. Dataset-Level Context Aggregation

As shown in Fig. 3, the memory module [28] is introduced to store information about the dataset-level distribution of various categories, that is, $\mathcal{M}_d$. For convenience of the presentation, we use the subscript $[i,\ j]$, $[i,\ :]$, or $[:,\ j]$ to index the single element or elements in a certain position, row, or column in the matrix. The feature representation R_n for each category is first split from the original feature representation R

$$R_n = \ \left\{ R^{\frac{HW}{64} \times C}_{[i,\ :]} | (\mathrm{GT}_i = n) \cap (i \subseteq HW) \right\} \tag{7}$$

where n $(1 \leq n \leq N)$ represents the corresponding category, i represents the ith pixel of the matrix R. GT denotes the category label of ground-truth.

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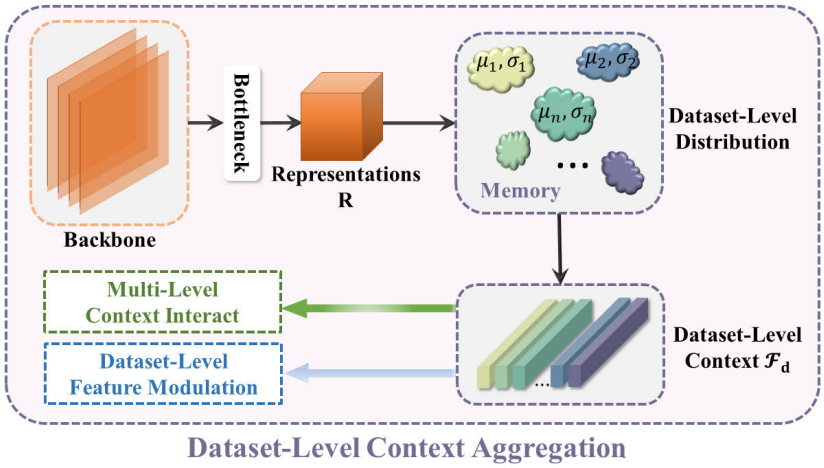


Fig. 3. Detailed illustration of our dataset-level context aggregation. The memory module is employed to store dataset-level contextual cue from historical image set. $\mathcal{F}_d$ is then used to enhance the performance of DLFM and MLCI, which is key to enable the consistency of different level contextual cues.

Subsequently, the category feature representation R_n is projected to a Gaussian distribution. This operation is inspired by the VAE, which has been verified that valid information is not lost during the reconstruction process. In addition, the advantage of this operation that it is simple yet efficient. For the dataset-level distribution information of each category, only two hyperparameters, mean and variance, are stored and updated to achieve accurate modeling. In other words, the memory module's storage is only an $N \times 2$ matrix. Afterward, these hyperparameters are updated during the training iterations to refine the dataset-level high-dimensional contextual space representation. The update strategy of hyperparameters is as follows:

$$\begin{aligned} \mu_{\omega_n} &= \beta \cdot \mu_{\omega_n - 1} + (1-\beta) \cdot \text{Mean}\left(R_{\omega_n}\right) \\ \sigma_{\omega_n} &= \beta \cdot \sigma_{\omega_n - 1} + (1-\beta) \cdot \text{Std}\left(R_{\omega_n}\right) \end{aligned} \tag{8}$$

where ω denotes the current number of iterations; μ and σ are the mean and standard deviation; and β is an artificial hyperparameter (empirically set to 0.1), preventing outliers in individual samples that cause the summary distribution to deviate significantly from the true distribution. The parameters after the last iteration are stored

$$M = \begin{bmatrix} \mu_1 & \sigma_1 \\ \vdots & \vdots \\ \mu_N & \sigma_N \end{bmatrix} = \begin{bmatrix} \mu_{\omega_1} & \sigma_{\omega_1} \\ \vdots & \vdots \\ \mu_{\omega_N} & \sigma_{\omega_N} \end{bmatrix} \tag{9}$$

where the size of M is $N \times 2$. Finally, the dataset-level distribution information is calculated as follows:

$$\mathcal{F}_d = \mathcal{M}_c\left(M_{[:,\ 0]},\ M_{[:,\ 1]}\right) \tag{10}$$

where $M_{[:,\ 0]}$ and $M_{[:,\ 1]}$ are tensors with the mean and standard deviation, respectively, of the distribution information for each class. These tensors provide valuable insights into the statistical characteristics of dataset level. Then, $\mathcal{M}_c$ returns a tensor of random numbers drawn from separate normal distributions whose mean and standard deviation are given. Finally, a high-dimensional dataset-level contextual cue $\mathcal{F}_d$ is aggregated, which plays a pivotal role in subsequent DLFM and MLCI.

C. Dataset-Level Feature Modulation

Here, as shown in Fig. 4, we elaborate the DLFM $\mathcal{J}$, which aims to adaptively enhance the semantic information of the original feature representations by leveraging the dataset-level contextual cue $\mathcal{F}_d$. To begin, a classification head $\mathcal{W}$ is first utilized to predict the category probability distribution Wights_R of the representations R

$$\text{Wights}_R = \mathcal{W}(R) \tag{11}$$

where $\mathcal{W}$ is implemented by $1 \times 1\text{Conv} \rightarrow \text{BN} \rightarrow \text{ReLU} \rightarrow \text{Dropout} \rightarrow 1 \times 1\text{Conv}$. The matrix size of Wights_R is $N \times (H/8) \times (W/8)$. Then the enhanced representation $\mathcal{F'}_{do}$ is calculated

$$\mathcal{F}'_{do} = \text{permute}\left(\text{Wights}_R\right) \otimes \mathcal{F}_d \tag{12}$$

where permute is a transformation function used to adjust Wights_R to have size of $(HW/64) \times N$. $\otimes$ stands for matrix multiplication. After that, to further recalibrate $\mathcal{F}'_{do}$, we calculate the relation between $\mathcal{F}'_{do}$ and R to obtain a detail confidence weight Distance_{do}

$$\begin{aligned} R_{dis}^{\frac{HW}{64} \times \frac{C}{2}} &= \text{permute}\left(\text{Conv}\left(R^{C \times \frac{H}{8} \times \frac{W}{8}}\right)\right)^{\text{T}} \\ \mathcal{F}_{do_{\text{dis}}}^{\frac{C}{2} \times \frac{HW}{64}} &= \text{permute}\left(\text{Conv}\left(\mathcal{F}_{do}^{\prime C \times \frac{H}{8} \times \frac{W}{8}}\right)\right) \end{aligned} \tag{13}$$

$$\text{Distance}_{do}^{\frac{HW}{64} \times \frac{HW}{64}} = \text{Softmax}\left(\frac{R_{dis}^{\frac{HW}{64} \times \frac{C}{2}} \otimes \mathcal{F}_{do_{dis}}^{\frac{C}{2} \times \frac{HW}{64}}}{\sqrt{\frac{C}{2}}}\right) \tag{14}$$

where Conv is a 1×1 convolutional layer used to adjust the dimension. And permute is used to resize the matrix. For convenience, superscripts are used to denote the size of matrix. Finally, the detail confidence weights Distance_{do} are integrated into $\mathcal{F}_{do}$

$$\mathcal{V}^{\frac{HW}{64} \times \frac{C}{2}} = \left(\text{permute}\left(\text{Conv}\left(\mathcal{F}_{do}^{C \times \frac{H}{8} \times \frac{W}{8}}\right)\right)\right)^{\text{T}} \tag{15}$$

$$\mathcal{F}_{do}^{C \times \frac{H}{8} \times \frac{W}{8}} = \text{permute}\left(\text{Conv}\left(\text{Distance}_{do}^{\frac{HW}{64} \times \frac{HW}{64}} \otimes \mathcal{V}^{\frac{HW}{64} \times \frac{C}{2}}\right)\right). \tag{16}$$

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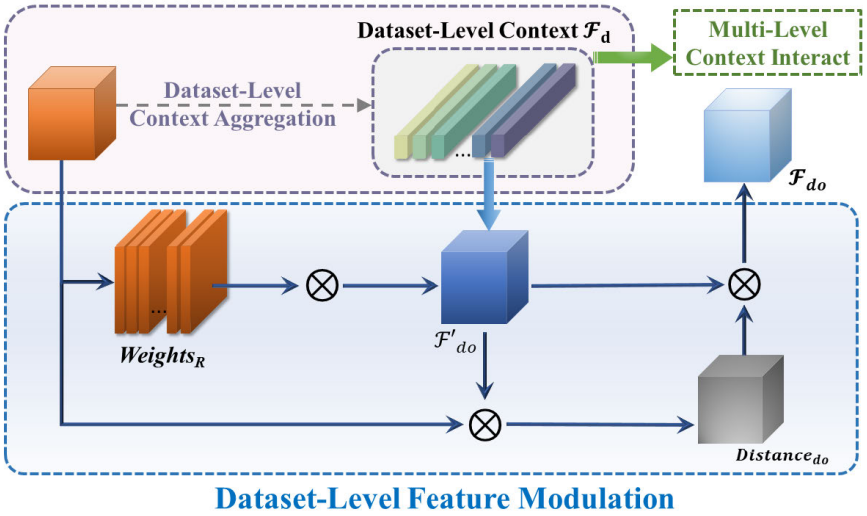


Fig. 4. Illustration of the pipeline of our DLFM. DLFM is designed to incorporate the yielded dataset-level contextual cue into the original representations R. This modulation with detail confidence weight Distance_{do} ensures that the output $\mathcal{F}_{do}$ is refined and reliable.

Similarly, Conv is a dimension adjustment operation based on the convolutional layer. Finally, the output of DLFM, $\mathcal{F}_{do}$ of size $C \times (H/8) \times (W/8)$, is what we demand.

D. Multilevel Context Interaction

As shown in Fig. 5, the MLCI $\mathcal{G}$ is designed for integrating the complementary dataset-level contextual cue $\mathcal{F}_d$ with the image-level contextual cue. Firstly, $\mathcal{F}_d$ is utilized to predict the category probability distribution $\mathcal{F}'_d$ of the representation R

$$\mathcal{F}'_d = \mathcal{F}_d \otimes R \tag{17}$$

where the size of $\mathcal{F}'_d$ is $N \times (H/8) \times (W/8)$, while the decoder $\mathcal{I}$ is utilized to initially model the image-level context $\mathcal{F}_i$

$$\mathcal{F}_i = \mathcal{I}(R) \tag{18}$$

where the matrix size of $\mathcal{F}_i$ is $C \times (H/8) \times (W/8)$. Specifically, the decoder $\mathcal{I}$ is borrowed from UperNet [59]. Note that, the initial image-level context decoder can be replaced by all of the existing methods for better modeling the image-level contextual cue when pursuing the best segmentation performance and computation complexity. It does not affect the basic motivation of this article.

After that, according to the category probability distribution $\mathcal{F}'_d$, the initial image-level context $\mathcal{F}_i$ can be grouped into different category regions

$$\mathcal{F}'_{di} = \mathcal{G}'\left(\mathcal{F}'_d, \mathcal{F}_i\right) \tag{19}$$

where the size of $\mathcal{F}'_{di}$ is $(H/8) \times (W/8) \times C$. And $\mathcal{G}'$ is considered as the key of MLCI $\mathcal{G}$. In this process, category probability distribution $\mathcal{F}'_d$, viewed from the view of dataset, is exploited as a mediator. It facilitates the secondary aggregation of image-level contextual cue, enabling well-structured and consistent interaction of multilevel contextual cues. The pseudo code of $\mathcal{G}'$ is shown in Algorithm 1.

Algorithm 1 Pseudo Code of Generator $\mathcal{G}'$

```
F_d = F_d.reshape(num_channels, −1).permute(1, 0)
F_i = F_i.reshape(num_channels, −1).permute(1, 0)
# Group into category regions according to Dataset-Level
argmax = F_d.argmax(1)
# Interaction by category
for n in range(N):
    mask = (argmax == n)
    F_i_cls = F_i[mask]
    F_d_cls = F_d[:, n][mask]
    weight = softmax(F_d_cls, dim = 0)
    F_i_cls = F_i_cls * weight.unsqueeze(−1)
    F_i_cls = F_i_cls.sum(0)
    F_i[mask] = F_i_cls
F_i = F_i.reshape(h, w, num_channels)
```

Then, the transform function permute is utilized for integration of $\mathcal{F}'_d$

$$\mathcal{F}'_{di} = \text{permute}\left(\mathcal{F}'_d\right) \tag{20}$$

where the size of $\mathcal{F}'_{di}$ is $C \times (H/8) \times (W/8)$. After that, to further recalibrate $\mathcal{F}'_{di}$, we calculate the correlation between $\mathcal{F}_i$ and $\mathcal{F}'_{di}$ to obtain the regional confidence weights Distance_{di}

$$\begin{aligned} X &= \text{permuteConv}(\mathcal{F}_i) \\ Y &= \text{permuteConv}\left(\mathcal{F}'_{di}\right) \end{aligned} \tag{21}$$

$$\text{Distance}_{di} = \text{Softmax}\left(\frac{X^{\mathrm{T}} \otimes Y}{\sqrt{\frac{C}{2}}}\right). \tag{22}$$

Finally, the regional confidence weights Distance_{di} are integrated into $\mathcal{F}'_{di}$

$$\mathcal{F}_{di} = \text{permute}(\text{Conv}(\text{Distance}_{di} \otimes Y)). \tag{23}$$

Although the above operation is similar to the last step of $\mathcal{F}_{do}$ (C) from the view of computational formula. However, it should be noted that, due to the different sources of

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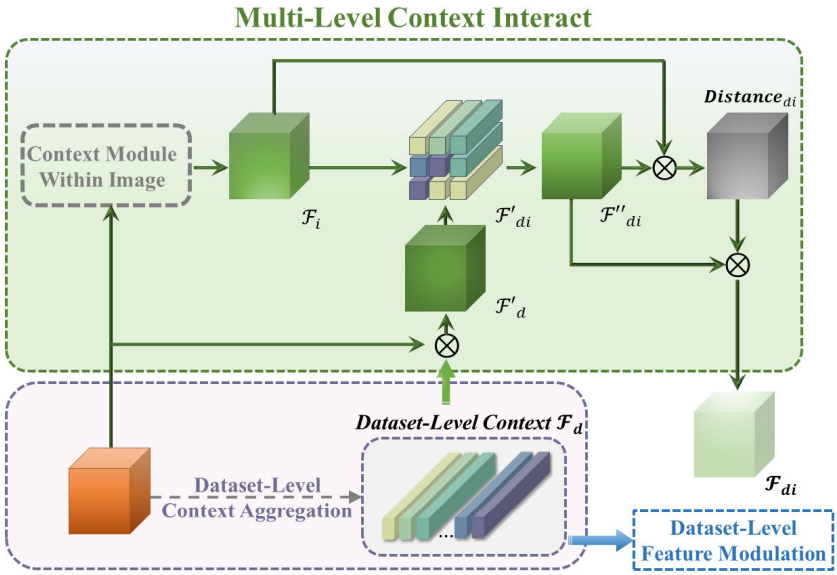


Fig. 5. Detailed illustration of our MLCI. The dataset-level contextual cue is utilized to assist the fusion of image-level context cues. With regional confidence weight Distance_{di}, the conflicting information accompanying the interaction process is minimized. The "Context Module Within Image" denotes for applying existing image-level context decoder (e.g., UperNet [59]), which is an optional operation.

weights for quantitative analysis, the detail confidence weight Distance_{do} is used to enhance the refined pixel representation of the output of DLFM. In contrast, the regional confidence weight Distance_{di} is used to calibrate the $\mathcal{F}'_{di}$ in a secondary way to minimize the conflicting information accompanying the interaction process and achieve the well-structured integration.

E. PFI-Net Based on CST

As shown in Fig. 6, for the parallel contextual cues $\mathcal{F}_{do}$, $\mathcal{F}_{di}$, we have already elaborated the general interaction process of PFI-Net in A. In this section, we focus on the internal construction of the CST block and a proposed multihead cross-attention mechanism based on shifted windows, swin-window multihead cross attention (SW-MCA).

The CST block aims to calculate the relevance weights between cue x and cue y and incorporate them into a new contextual cue with cross-attention mechanism. As depicted in Fig. 6(a), during an interaction, x and y undergo encoding via the embedding layer. Since the input features have already interacted in the contextual aggregation scheme, there is no great span of extensive correspondence between the contextual cues. Then we divide the feature maps of these contextual cues into two sets of windows: the basic window set and the search window set. These sets have different window partitions and compute the cross-attention weights within their respective windows. The basic window set is divided into non-overlapping pixels of fixed size, which is referred to as the window partition layer. The search window set is partitioned as shown in Fig. 6(b), to compensate for the absence of a window-based attention mechanism for modeling long-distance dependencies, we introduce the shifted-window partitioning method from the swin transformer [19]. This method introduces cross-window connections while still effectively computing non-overlapping windows, which is referred to as the shifted window partition Layer.

After that, as in Fig. 6(c), based on these window sets, we proposed SW-MCA to capture the mutual correspondence between the basic window and the search window by calculating their relevance weights. Moreover, it also employs multihead attention to fully represent subspaces. SW-MCA is essentially a mapping function that maps query and key-value pairs to the output. Specifically, the dot product of query and key is computed following [19], [60], [61], [62] and relative position bias is introduced. After that, the softmax function is utilized to obtain the cross-attention weights, and finally, the weighted sum of values is computed to obtain the output

$$\text{SW}-\text{MCA}(Q_b, K_s, V_s) = \text{Softmax}\left(\frac{Q_b \otimes K_s}{\sqrt{d}} + B\right) \otimes V_s \tag{24}$$

where Q_b is the query from the base window; K_s, V_s are the key and value from the search window; d is the dimension of the query and key; $\otimes$ is the matrix multiplication; and B is the relative position bias introduced. After that, the output of SW-MCA is fed into the MLP layer [63], [64], [65], [66] to enhance the learning capability, while LayerNorm (LN) layer is applied before each SW-CST and MLP to ensure the effectiveness of each layer, and residuals are introduced after each SW-CST and MLP layer for optimizing the training.

Finally, two of the above CST blocks are connected in parallel to form a subnetwork, and three layers of subnetwork are cascaded to form a PFI-Net. The multilevel contextual cues $\mathcal{F}_{do}$ and $\mathcal{F}_{di}$ are gradually fused and matched in multiple subnetworks, thus achieving a fine-grained interaction.

It is also worth stating that our CST block was originally inspired by the swin transformer framework. However, depending on the specific scene and task, we make some improvements to the conventional approach.

1) The input feature map size of CST is much smaller than the original image input. We eliminate the patch slicing

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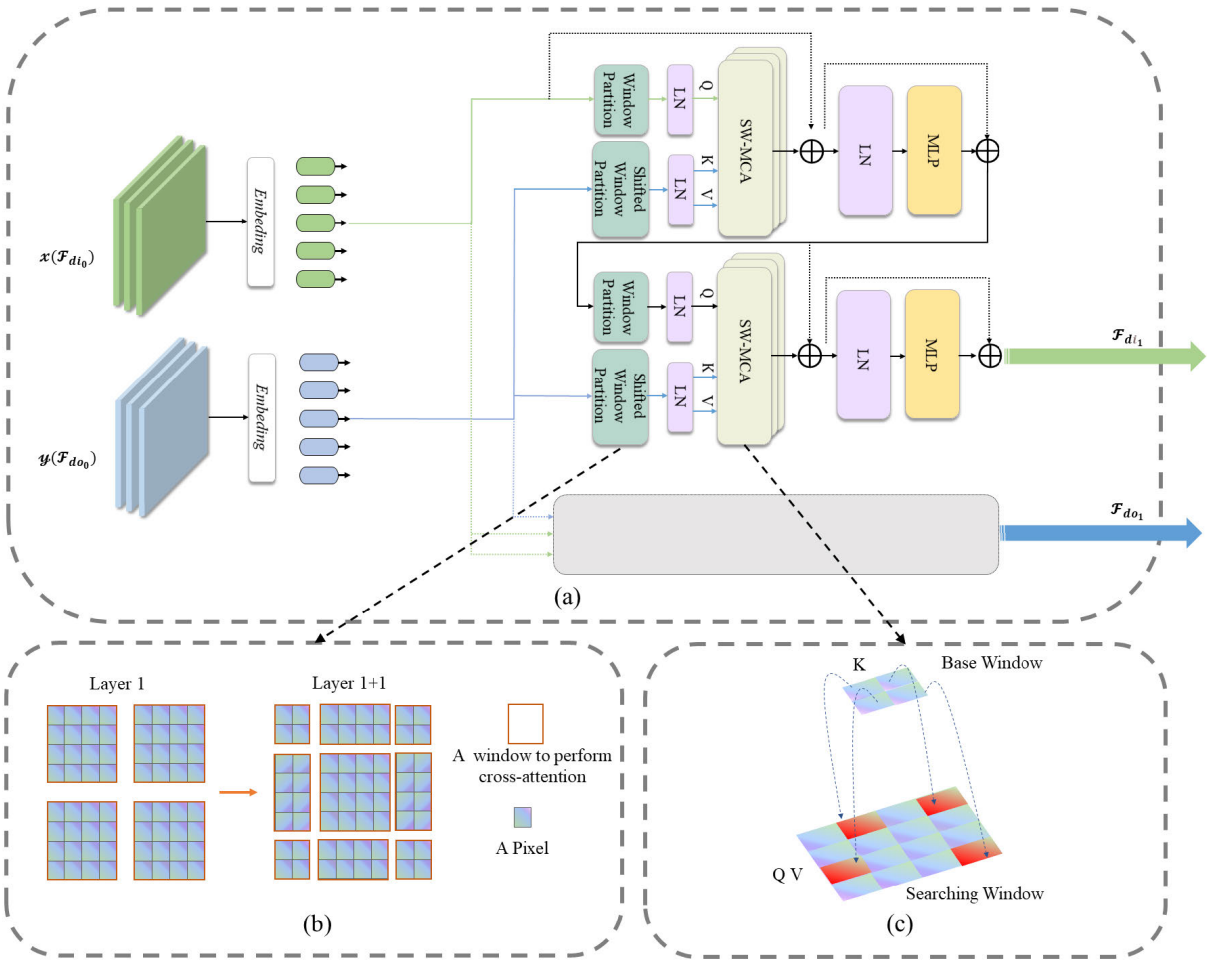


Fig. 6. Detailed illustration of our PFI-Net based on CST. (a) Construction of a pair of CST blocks, the gray areas are constructed the same as the two specifically drawn except for the inputs. A subnetwork in Fig. 2 consists of two CST blocks. (b) Shifted Window Partition, we have two different methods of window partition to generate basic and search windows. (c) We compute the cross attention between basic and search windows by SW-MCA.

operation in the conventional network, and directly model the relationship between feature pixels, so as to learn the correspondences between pixels and facilitate the features to be automatically fused.

2) Compared with the conventional network (e.g., Swin-T [19], 12 layers), we limit the number of layers in each CST block to two. This is because the input features of the CST block already incorporate deep features and contextual information from different levels. For this reason, two cross-attention layers are sufficient to accomplish the interaction while improving computational efficiency.
3) We eliminate the patch merging layer for multiscale extraction, so the CST block is end-to-end, which saves computational resources and has good generalization at the same time.

IV. Experiments

We conduct the experiments on three widely used semantic segmentation benchmarks, that is, Vaihingen dataset [13], GaoFen-2 dataset [33], and iSAID dataset [34].

A. Dataset Description

1) *Vaihingen Dataset:* The ISPRS 2-D semantic labeling Vaihingen [13] dataset is provided by the International Society for Photogrammetry and Remote Sensing, which consists of 33 high-resolution true orthophoto tiles and corresponding digital surface models, and the DSM band channel is abandoned in our experiment. The average image size is 2494 × 2064. Among the 33 tiles, 16 tiles are used for training and validation, specifically, 11 tiles are used for training and 5 tiles are for validation. The rest of the tiles is used for testing the performance of the method. The categories of the labeling include six landcover classes: impervious surface, building, low vegetation, tree, car, and clutter. We perform experiments on the whole six categories.
2) *GaoFen-2 Dataset:* GaoFen-2 [33] dataset is a more challenging dataset compared with the ISPRS 2-D Semantic Labeling Vaihingen data. It contains 500 satellite images collected from the GaoFen-2 satellite over different geographic locations in China, which is featured by high intra-class diversity and low inter-class separability. The GaoFen-2 is split into 400 training and 100 validation images with annotation containing nine

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TABLE I
ABLATION STUDY OF EMIDI ON VAIHINGEN, GAOFEN-2, AND ISAID. HERE "✓" INDICATES THAT THIS COMPONENT IS APPLIED

BL	DLFM	MLCI	CDS	FPN	PFI-Net	Params(M)	FLOPs(G)	mIoU on Vaihingen(%)	mIoU on GaoFen-2(%)	mIoU on iSAID(%)
✓						101.79	134.65	67.16	57.77	62.74
✓	✓					106.94	179.1	68.83	59.06	63.40
✓		✓				107.17	185.47	68.93	58.85	63.62
✓	✓	✓	✓			115.7	211.29	70.41	60.79	64.21
✓	✓	✓		✓		138.78	261.78	70.27	60.16	63.99
✓	✓	✓			✓	122.27	235.95	**71.5**	**62.49**	**66.48**

categories: road, building, tree, grass, bare land, water, transportation, impervious surfaces, and others.

3) *iSAID Dataset:* The instance segmentation in aerial images dataset (iSAID) [34] was modified from a large-scale object detection dataset DOTA [67], which fully represents the common feature and scale distribution differences in remote sensing images. This large-scale and densely annotated dataset contains 655 451 object instances for 15 categories across 2806 high-resolution images. The categories of the labeling include 15 landcover classes: ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, bridge, large vehicle, small vehicle, helicopter, swimming pool, roundabout, soccer ball field, plane, and harbor. The original image sizes of this dataset range from 800 × 800 pixels to 4000 × 13 000 pixels. These diverse remote sensing images and object instances make ISAID suitable for practical applications in complex aerial scenes. Because the semantic annotations of the test set are not available, the validation set was used to evaluate the performance of the proposed method and the reference methods.

B. Evaluation Metrics

Accuracy and efficiency are important metrics for the HRS small object semantic segmentation task. The intersection over union (IoU) is chosen for the evaluation accuracy, and the computation complexity is used for efficiency. The IoU is calculated as follows:

$$\mathrm{IoU}_i = \frac{x_{ii}}{\sum_{i=1}^{n} x_{ij} + \sum_{i=1}^{n} x_{ji} - x_{ii}} \tag{25}$$

$$\mathrm{mIoU} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{IoU}_i \tag{26}$$

where x_{ij} represents the number of instances of class i predicted as class j, and n is the number of classes.

For the model efficiency, the computational complexity is evaluated, that is, number of model parameters (Params) and floating point operations (FLOPs).

C. Training Settings

During the training, the AdamW optimizer is adopted with betas of (0.9, 0.999) and a weight decay of 0.01. The initial learning rate is set to 0.00006, and a "Poly" schedule with power 0.9 is applied. The training set images are uniformly scaled to 512 × 512 size, and the scaled images are enhanced by random mirror flip and Gaussian blur. All the networks are implemented using four RTX Titan GPUs with PyTorch deep learning framework. The backbone networks used by our method and comparative methods are pre-trained on ImageNet. The other training settings are slightly different due to the chosen dataset.

1) *Vaihingen Dataset/ GaoFen-2 Dataset:* For Vaihingen dataset and GaoFen-2 dataset, the number of training iterations is set to 25k and the batch size is set to 4. The original image of Vaihingen dataset is cropped and the size of the cropped image is 319 × 428 [33].
2) *iSAID Dataset:* For iSAID dataset, the number of training iterations is set to 100k and the batch size is set to 8. Due to the large size of each image, sliding window inference technology is adopted. We adapt the same dataset division as [5] and [68]; the size of the window patch is 896 × 896, and the stride is 512. For the overlapping pixels, the mean of the prediction probability is adopted.

D. Ablation Analysis

In this section, as shown in Table I, in order to evaluate the effectiveness and robustness of EMIDI, we have performed an ablation analysis of the different components of our network on the Vaihingen, GaoFen-2, and iSAID datasets. Specifically, these components include baseline (BL), DLFM module, MLCI module, and PFI-Net based on CST. In addition, we introduce two conventional feature fuse schemes for comparison, the channel dimensional splicing (CDS) and the feature pyramid network (FPN).

1) *BL:* The BL is used to assess the effectiveness of other modules in the network, only utilizing the context module within image (UperNet) after the backbone. The BL achieves mIoU of 67.16%, Params of 101.79 M, and FLOPs of 134.65 G on Vaihingen dataset.
2) *BL + DLFM:* Compared with the BL, we introduce the DLFM to adaptively bootstrap the feature modulation and enhance the contextual cues from original feature representation. The results show that our DLFM improves network performance significantly and efficiently. compared with BL, the mIoU improves by 1.67%–68.83%. The application of DLFM introduces only a small amount of computation complexity, yet it yields a significant improvement in accuracy.
3) *BL + MLCI:* We apply our MLCI to the BL, aiming at a well-structured integration of contexts and maximizing the conflicting information. The results show that our MLCI is effective, which significantly improves the performance of BL by 1.77% of mIoU to 68.93% while

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the Params and FLOPs are only increased to 107.17 M and 185.47 G.

4) *BL + DLFM + MLCI + CDS:* In this experiment, we integrate both DLFM and MLCI modules into the BL to evaluate the compatibility of our proposed two modules. After that, the contextual cues generated by these two modules are directly spliced along the dimension of feature channel. For convenience, we call this simplistic fusion as CDS. As shown in Table I, we can find that the segmentation performance of BL + DLFM + MLCI + CDS outperforms the BL by 3.25% in terms of mIoU to 70.41%. It is a much more significant improvement than the schemes that apply a single DLFM module (3.25% versus 1.67%) or a single MLCI module (3.25% versus 1.77%). This result shows that DLFM and MLCI can work together well to enable the contextual cues more effective and reliable, which demonstrates the good reliability and compatibility of our proposed modules.
5) *BL + DLFM + MLCI + FPN:* Due to its simplicity, CDS is a widely used scheme of feature fusion. However, contextual cues captured from multilevels without further interaction can also lead to pixel misclassification. Therefore, we try to introduce the widely used FPN module to replace CDS for feature interaction and fusion. The experimental results show that compared with BL + DLFM + MLCI + CDS, the introduction of FPN not only does not bring optimization, but also decreases the performance to 70.27% mIoU. In addition, the Params and FLOPs of FPN are 138.78 M and 261.78 G, which is the most "expensive" variant in all experiments. Obviously, BL + DLFM + MLCI + FPN is a wrong attempt in terms of feature fusion, and we need to explore a better scheme.
6) *BL+DLFM+MLCI+PFI-Net:* Aiming at the problem in (5), we propose a PFI-Net based on CST. The cross-attention mechanism with shifted windows is utilized to achieve fine-grained interaction with multilevel contextual cues. As shown in Table I, after replacing the interaction scheme with PFI-Net, the segmentation performance is improved by 1.09% compared with CDS, while the computational complexity is only increased by 6.57 M Params and 24.66 G FLOPs. Compared with FPN, the method with PFI-Net even saves 16.51 M Params and 25.83 G FLOPs while obtaining a 1.23% improvement of mIoU. The results of this experiment fully demonstrate the necessity of PFI-Net. In overall analysis, our proposed BL + DLFM + MLCI + PFI-Net network framework exhibits optimal performance with 71.5% mIoU, which is significantly higher than the BL 4.34% mIoU, and the Params is controlled in an acceptable range. These results well demonstrate the reliability, effectiveness, and efficiency of our proposed EMIDI.

In addition to Vaihingen, we further conduct ablation experiments on GaoFen-2 and iSAID datasets in Table I to investigate the robustness of each module. As shown, on different datasets, each module of our proposed EMIDI has again proved to be effective. Specifically, the integration of DLFM or MLCI significantly improves the performance of BL by 1.29%, 1.08% (on GaoFen-2) and 0.66%, 0.88% (on iSAID), respectively. Furthermore, we can find that combining DLFM and MLCI with PFI-Net outperforms all the other methods by yielding an mIoU of 62.49% (on GaoFen-2) and 66.48% (on iSAID). The improvement is much higher than applying CDS (4.72% versus 3.02% on GaoFen-2, 3.74% versus 1.47% on iSAID) or FPN (4.72% versus 2.39% on GaoFen-2, 3.74% versus 1.25% on iSAID). These results are consistent with the previous observation on Vaihingen dataset, and well demonstrate the robustness of our proposed modules.

To further demonstrate the advantages of each proposed module, as shown in Fig. 7, visual analysis is conducted. Due to the lack of interaction, the BL fails to preserve the details near the boundary, and generates incorrect labels. As exemplified in Row 1 of Fig. 7, the prediction labels of BL are cluttered. The visualization results of BL + DLFM and BL + MLCI, although bringing about an improvement in the performance metrics, still present a large number of misclassifications. Both of them cannot meet the needs of downstream tasks.

Afterward, the "collaboration" between the DLFM and MLCI modules improves the prediction results significantly. However, it is not as effective in edge details and complex samples, such as pixels with obscure or confusing features. These features are challenging to handle with current methods. For example, buildings obstructed by vehicle shadows (Row 2 of Fig. 7). Particularly, when applying FPN as a fusion scheme, the desired optimization of edge details through the incorporation of multiscale information does not occur. In fact, the "hard" contextual fusion engenders a substantial amount of conflicting information, exacerbating the issue of pixel obfuscation.

The proposed EMIDI is designed to solve the above problems by ensuring contextual consistency and avoiding conflicting information. The BL + DLFM + MLCI + PFI-Net variant outperforms the BL by achieving optimal performance. The visualization results of this method are significantly superior to previous variants. In a complex interlaced building scene, Row 3 of Fig. 7 demonstrates the fine segmentation of edges achieved by this method. In Fig. 7, Rows 2 and 4 demonstrate the method's accurate judgment on complex samples affected by shadows and other disturbing information.

E. Comparison With the State-of-the-Art Semantic Segmentation Methods

In this section, we compare the performance of the proposed EMIDI model with state-of-the-art semantic segmentation methods on three publicly available remote sensing datasets: Vaihingen, GaoFen-2, and iSAID. These representative methods include: U-Net [47], PSPNet [51], DeeplabV3+ [25], UperNet [59], DANet [69], OCRNet [21], Segmenter [26], ISNet [27], MCIBI++ [22], CMTFNet [70], RSSFormer [5], IDRNet [71], BGCANet [72]. The segmentation accuracies using mean IoU (mIoU) and IoU per category are reported as shown as Tables II, III, and V.

1) *Results on Vaihingen:* Table II presents a comparison of the segmentation results for the validation set

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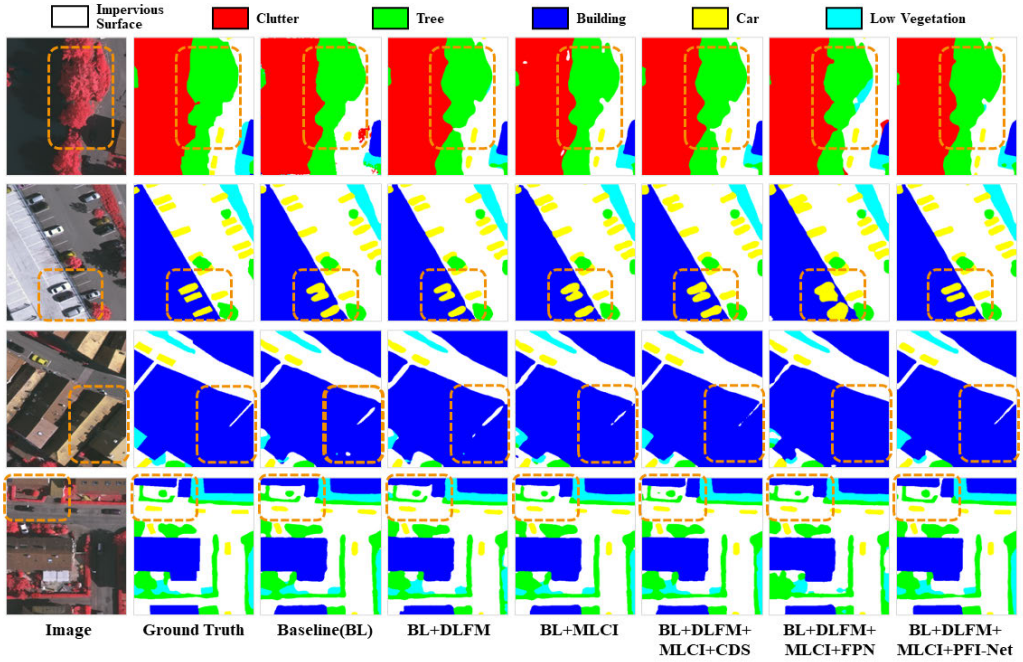


Fig. 7. Visualized comparison for ablation study on Vaihingen dataset. (Left to Right) Input image, the ground-truth, and the results from variants of our methods are shown.

TABLE II

COMPARISON WITH THE STATE-OF-THE-ART METHODS ON VAIHINGEN DATASET. WE REPORT IOU FOR EVERY CATEGORY, OVERALL MIOU. THE BOLD VALUES IN EACH COLUMN INDICATE THE BEST ENTRIES WITH BACKBONE OF RESNET101

Method	Backbone	IoU per category (%)						mIoU(%)
		Imp. surf	Clutter	Tree	Building	Car	Low veg.	
U-Net$_{MICCAI'2015}$ [47]	ResNet101	76.41	15.16	73.58	82.55	47.44	63.41	59.76
PSPNet$_{CVPR'2017}$ [51]	ResNet101	78.45	29.77	75.18	84.44	54.66	63.5	64.33
DeepLabV3+$_{ECCV'2018}$ [25]	ResNet101	80.17	41.63	75.49	86.64	65.21	66.03	69.19
UperNet$_{ECCV'2018}$ [59]	ResNet101	79.62	39.65	75.46	86.41	64.1	65.19	68.4
DANet$_{CVPR'2019}$ [69]	ResNet101	79.9	38.79	75.14	86.73	63.26	65.31	68.19
OCRNet$_{ECCV'2020}$ [21]	HRNetV2-W48	80.04	39.31	75.07	86.73	63.62	65.23	68.33
Segmenter$_{CVPR'2021}$ [26]	ViT-L	79.67	39.47	75.39	86.47	63.92	65.38	68.38
ISNet$_{ICCV'2021}$ [27]	ResNet101	79.51	42.41	75.43	86.63	64.9	65.35	69.04
MCIBI++$_{TPAMI'2022}$ [22]	ResNet101	80.36	45.09	**75.78**	88.24	66.29	66.51	70.38
CMTFNet$_{TGRS'2023}$ [70]	ResNet101	80.21	43.54	75.63	86.98	63.38	65.73	69.25
RSSFormer$_{TGRS'2023}$ [5]	RSS-B	80.5	44.11	75.61	87.55	67.22	65.48	70.07
IDRNet$_{NeurIPS'2023}$ [71]	ResNet101	80.48	37.49	75.43	87.51	64.61	65.34	68.98
BGCANet$_{TIP'2023}$ [72]	ResNet101	80.75	41.87	76.47	86.05	67.00	65.19	69.09
EMIDI(Ours)	ResNet101	**80.94**	**49.58**	75.72	**88.48**	**67.5**	**66.74**	**71.5**
EMIDI(Ours)	Swin-L	80.41	52.04	75.38	89.93	67.27	67.05	72.01

of Vaihingen when using the ResNet101 backbone. MCIBI++ achieves an mIoU of 70.6% by utilizing dataset-level contexts for semantic segmentation, which surpasses any previous best method. In comparison to MCIBI++, which employs a single dataset-level context, the utilization of well-designed multilevel contextual cues in our proposed EMIDI model has led to a significant enhancement in performance. Specifically, our EMIDI method, leveraging the same ResNet101 backbone, achieves superior performance with a 71.5% mIoU. This outcome serves as validation for the effectiveness of our proposed method. Additionally, we include a visualization of a subset of representative segmentation results in Fig. 8, highlighting the exceptional capabilities of MCIBI in complex sample processing.

2) *Results on GaoFen-2:* To further validate the effectiveness of our approach, we extend the application of EMIDI to a more challenging remote sensing segmentation benchmark known as GaoFen-2. The performance and visualization of our EMIDI model on the GaoFen-2 dataset are presented in Table III and Fig. 9. Compared with the state-of-the-art models, our model achieves an mIoU of 62.49%. In terms of individual category IoU comparison, EMIDI surpasses all benchmarked methods, particularly in handling complex and perplexing samples, such as building and impervious surface, as well as trees and grass. This superior performance of EMIDI is attributed to its integration of the detail-biased image-level contextual cues. Notably, the GaoFen-2 training set comprises only a small number of images containing samples of bare land and water, while the remaining images are intentionally allocated to the validation set to evaluate the model's generalization ability. This distribution effectively shows the exceptional generalization capacity of EMIDI.

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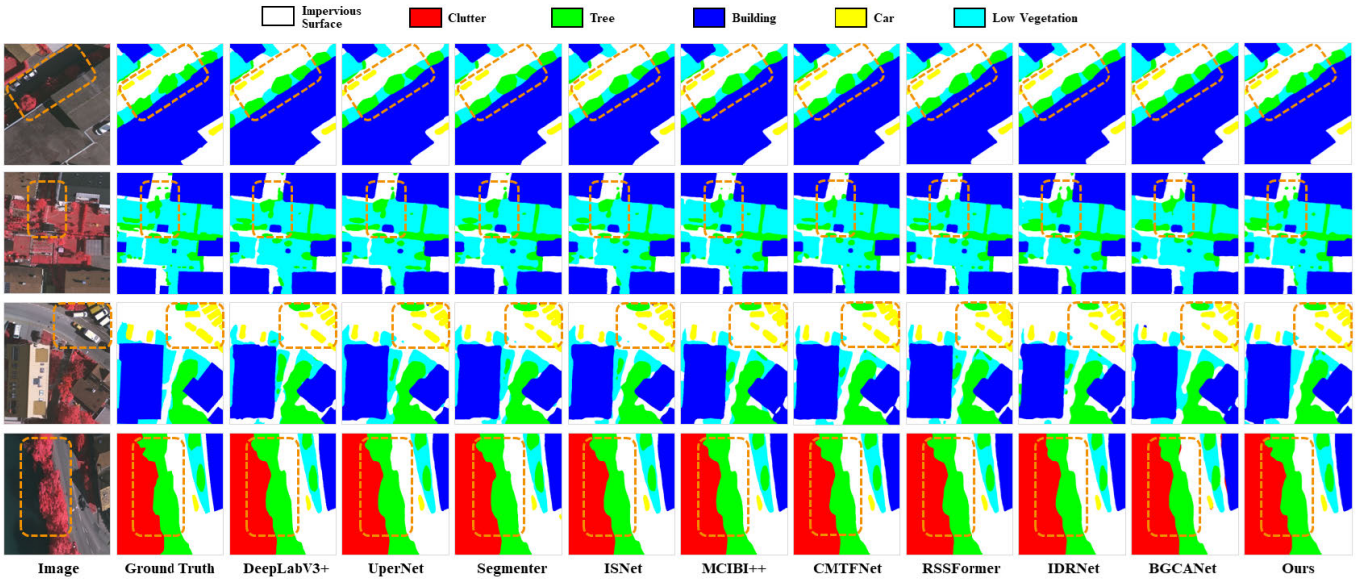


Fig. 8. Visualized comparison on Vaihingen dataset. (Left to Right) Input image, the ground-truth, the results from the State-of-the-art Methods and our methods.

TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON GAOFEN-2 DATASET

Method	Backbone	IoU per category (%)									mIoU(%)
		Imp. surf	Tree	Grass	Road	Building	Transportation	Bare Land	Water	Others	
U-Net$_{MICCAI'2015}$ [47]	ResNet101	45.98	79.38	64.65	71.39	68.47	45.43	44.81	0	0.4	46.72
PSPNet$_{CVPR'2017}$ [51]	ResNet101	47.61	80.72	68.53	72.34	70.96	44.02	44.5	14.3	7.59	50.16
DeepLabV3+$_{ECCV'2018}$ [25]	ResNet101	48.18	80.81	69.66	71.24	70.48	48.19	45.91	72.38	6.94	57.18
UperNet$_{ECCV'2018}$ [59]	ResNet101	47.64	81.61	68.24	70.99	71.31	48.17	44.29	80.9	6.76	57.76
DANet$_{CVPR'2019}$ [69]	ResNet101	46.55	80.88	69.15	72.75	68.76	48.1	43.9	45.76	7.86	53.75
OCRNet$_{ECCV'2020}$ [21]	HRNetV2-W48	46.35	82.25	67.83	73.82	69.76	47.57	38.06	59.9	8.47	54.89
Segmenter$_{CVPR'2021}$ [26]	ViT-L	48.5	81.44	68.78	70.94	72.94	48.13	43.49	77.62	7.14	57.66
ISNet$_{ICCV'2021}$ [27]	ResNet101	49.94	81.56	68.69	71.35	73.2	47.57	44.65	84.34	9.75	59.01
MCIBI++$_{TPAMI'2022}$ [22]	ResNet101	55.43	81.26	71.64	77.31	82.08	40.22	43.81	83.69	12.24	60.85
CMTFNet$_{TGRS'2023}$ [70]	ResNet101	53.32	81.37	68.37	74.81	73.73	**57.25**	36.96	80.55	10.03	60.71
RSSFormer$_{TGRS'2023}$ [5]	RSS-B	51.69	80.63	68.75	74.56	73.11	54.03	37.99	67.87	4.65	58.37
IDRNet$_{NeurIPS'2023}$ [71]	ResNet101	50.02	82.17	68.76	73.02	71.18	47.63	43.66	72.65	11.92	59.89
BGCANet$_{TIP'2023}$ [72]	ResNet101	47.95	**82.33**	**72.41**	69.89	68.75	57.16	43.71	**91.09**	10.36	60.41
EMIDI(Ours)	ResNet101	**55.5**	81.8	72.04	**77.41**	**83**	42.78	**46.6**	89.02	**14.32**	**62.49**
EMIDI(Ours)	Swin-L	56.93	82.5	71.67	77.39	83.41	42.78	50.72	90.8	12.87	63.23

TABLE IV

COMPARISON WITH STATE-OF-THE-ART METHODS ON ISAID DATASET

Method	Backbone	IoU per category (%)															mIoU(%)
		BG	Ship	ST	BD	TC	BC	GTF	Bridge	LV	SV	HC	SP	RA	SF	Plane	Harbor	
U-Net$_{MICCAI'2015}$ [47]	ResNet101	97.2	46.49	55.98	12.26	63.44	51.03	23.92	0	43.88	26.76	0	3.74	8.77	31.09	54.52	36.06	34.7
PSPNet$_{CVPR'2017}$ [51]	ResNet101	98.73	59.71	66.27	62.98	82.05	52.11	53.45	31.67	56.61	42.74	29.65	39.89	67.09	72.98	75.41	53.82	59.07
DeepLabV3+$_{ECCV'2018}$ [25]	ResNet101	98.82	62.25	67.93	75.84	85.21	60.6	51.14	42.88	58.95	42.85	32.27	46.08	61.32	73.33	76.73	53.07	61.82
UperNet$_{ECCV'2018}$ [59]	ResNet101	98.96	65.45	71.62	47.54	87.51	60.35	61.5	42.88	60.01	42.45	36.5	43.91	73.22	76.03	81.1	54.1	62.7
DANet$_{CVPR'2019}$ [69]	ResNet101	98.87	63.04	66.11	69.41	86.15	46.83	60.21	35.35	58.06	41.31	37.18	45.7	56.37	76.1	77.23	56.04	60.87
OCRNet$_{ECCV'2020}$ [21]	HRNetV2-W48	98.94	63.72	70.02	73.76	87.69	52.18	59.76	40.84	59.26	42.99	21.91	46.2	75.04	74.28	78.88	53.67	62.45
Segmenter$_{CVPR'2021}$ [26]	ViT-L	98.83	65.08	70.49	75.62	87.87	52.45	52.41	40.78	55.84	39.15	32.32	46.22	71.13	61.75	79.55	54	61.47
ISNet$_{ICCV'2021}$ [27]	ResNet101	98.93	65.11	72.1	72.77	87.06	64.56	50.46	37.67	58.81	40.9	36.68	47.47	77.32	75.48	79.98	52.04	63.59
MCIBI++$_{TPAMI'2022}$ [22]	ResNet101	98.99	66.13	71.57	79.24	87.32	55.25	59.29	41.69	57.42	38.92	38.77	47.32	73.06	74.79	80.81	54.87	64.09
CMTFNet$_{TGRS'2023}$ [70]	ResNet101	99.03	65.02	71.67	77.01	87.62	61.37	**62.71**	40.19	59.98	43.68	16.94	47.35	72.74	76.71	80.48	55.49	63.63
RSSFormer$_{TGRS'2023}$ [5]	RSS-B	98.97	65.91	**73.55**	79.68	86.92	64.41	61.13	**44.76**	60.67	41.34	35.08	50.22	**78.28**	78.57	73.64	55.6	65.55
IDRNet$_{NeurIPS'2023}$ [71]	ResNet101	98.97	65.56	69.01	74.8	86.35	58.4	56.39	35.18	60.75	39.56	33.81	46.05	69.78	72.16	81.36	55.27	62.4
BGCANet$_{TIP'2023}$ [72]	ResNet101	**99.08**	66.53	72.81	76.05	88.45	60.24	57.04	39.02	60.44	32.52	38.05	47.46	74.26	76.07	80.24	55.3	64.61
EMIDI(Ours)	ResNet101	99.05	**66.95**	71.45	**80.92**	**88.57**	**65.02**	62	42.38	**61.11**	**43.69**	**39.4**	**50.37**	75.8	**79.3**	**81.44**	**56.2**	**66.48**
EMIDI(Ours)	Swin-L	99.11	67.05	72.99	80.1	88.53	64.72	62.56	44.33	61.17	45.48	39.44	51.59	77.83	77.62	82.19	58.34	**67.07**

The abbreviations are as follows: BG - background, ST - storage tank, BD - baseball diamond, TC - tennis court, BC - basketball court, GTF - ground track field, LV - large vehicle, SV - small vehicle, HC - helicopter, SP - swimming pool, RA - roundabout, SBF - soccer ball field.

3) *Results on iSAID:* In contrast to the Vaihingen and GaoFen-2 datasets, iSAID serves as a large-scale, fine-grained semantic segmentation benchmark in remote sensing, presenting additional challenges such as complex scale diversity and segmentation of small samples at extreme foreground and background scales. As a result, the absence of an interaction between multilevel contexts is prone to interference from conflicting information, often resulting in misclassification. These challenges have a significant impact on the performance of methods

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TABLE V
COMPLEXITY COMPARISON WITH STATE-OF-THE-ART METHODS. WE REPORT MIOU, PARAMS AND FLOPS FOR EVERY METHODS. EXCEPT FOR MIOU, THE SMALLER VALUES INDICATE THE BETTER PERFORMANCE

Method	Backbone	mIoU on Vaihingen(%)	mIoU on GaoFen-2(%)	mIoU on iSAID(%)	Params(M)	FLOPs(G)
U-Net$_{MICCAI'2015}$ [47]	ResNet101	59.76	46.72	34.7	65.15	219.62
PSPNet$_{CVPR'2017}$ [51]	ResNet101	64.33	50.16	59.07	94.32	128.5
DeepLabV3+$_{ECCV'2018}$ [25]	ResNet101	69.19	57.18	61.82	81.35	138.63
UperNet$_{ECCV'2018}$ [59]	ResNet101	68.4	57.76	62.7	101.47	134.65
DANet$_{CVPR'2019}$ [69]	ResNet101	68.19	53.75	60.87	104.63	157.7
OCRNet$_{ECCV'2020}$ [21]	HRNetV2-W48	68.33	54.89	62.45	84.99	122.94
Segmenter$^{*}_{CVPR'2021}$ [26]	ViT-L	68.38	57.66	61.47	224.57	345.23
ISNet$_{ICCV'2021}$ [27]	ResNet101	69.04	59.01	63.59	95.42	180.6
MCIBI++$_{TPAMI'2022}$ [22]	ResNet101	70.38	60.85	64.09	128.65	210.59
CMTFNet$^{*}_{TGRS'2023}$ [70]	ResNet101	69.25	60.71	63.63	116.07	300.57
RSSFormer$^{*}_{TGRS'2023}$ [5]	RSS-B	70.07	58.37	65.55	149.23	483.21
IDRNet$_{NeurIPS'2023}$ [71]	ResNet101	68.98	59.89	62.4	108.65	349.23
BGCANet$_{TIP'2023}$ [72]	ResNet101	69.09	60.41	64.61	287.42	167.8
EMIDI(Ours)*	ResNet101	**71.5**	**62.49**	**66.48**	122.27	235.95
EMIDI(Ours)*	Swin-L	**72.01**	**63.23**	**67.07**	259.46	418.81

"*": Transformer-based method

such as IDRNet [71], BGCANet [72], CMTFNet [70], which generally excel in other datasets. Nevertheless, as indicated in Table IV, RSSFormer [5] achieves an mIoU of 65.55%. However, the EMIDI model surpasses the Transformer-based RSSFormer in various aspects. This superiority is attributed to EMIDI's integration of convolutional network for detailed information extraction and transformer for modeling multilevel contextual intrinsic correlation. The synergistic integration of these approaches in EMIDI results in enhanced performance, especially in addressing the intricate challenges posed by the iSAID dataset.

Specifically, the EMIDI of the ResNet-101 backbone significantly outperforms the previous best RSSFormer by 0.93% reaching 66.48% mIoU. To the best of our knowledge, it is the current optimal performance on iSAID benchmark. Meanwhile, combined with the visual analysis in Fig. 10, DLFM helps our model to achieve an advantage in terms of overall information, for example, up to 79.3% mIoU for a single class of soccerball field (SF) that usually cannot be accommodated in a single image. Furthermore, MLCIs provide additional small-sample processing advantages for EMIDI, for example, a 66.95% IoU for the ship class, which is usually poorly characterized, outperforming all existing methods. Finally, with this overall architecture that encourages MLCIs, EMIDI demonstrates significantly superior segmentation performance for complex samples that are easily confused under the interference of conflicting information, for example, the plane with a complex shape and hard-to-control edges has a single-class mIoU of 50.37%, and the harbor with an extreme aspect ratio has 88.57% IoU, both of which show the optimal performance.

Furthermore, to validate the proposed model's generalizability, we conduct an analysis of generalizability by changing the method's backbone. As shown in Tables II, III, and V, across three datasets, replacing the model's backbone from ResNet101 to Swin-L led to mIoU improvements of 0.51%, 0.74%, and 0.59% for each dataset. These improvements well demonstrate the model's generalizability and scalability. However, it should be noted that these improvements are primarily attributed to the superior quality of the employed backbone network. Consequently, this factor was not specifically emphasized in the comparative experiments to retain a focus on the advantages of the proposed model itself.

The comprehensive experimental results presented in this study clearly demonstrate the effectiveness of our proposed EMIDI method in the semantic segmentation of remote sensing images. It achieves a balance between computation complexity and accuracy, outperforming existing methods. Additionally, these results consistently support the main motivation of this article, which emphasizes the importance of promoting mutual interaction between dataset-level and image-level context for achieving fine-grained semantic segmentation in remote sensing images.

F. Complexity Comparison

Table V demonstrates the analysis of complexity comparison with existing methods, including the parameters, computation complexity (measured by the number of FLOPs). These metrics are calculated under the same settings during inference. "Params" refers to the number of parameters in the whole complete model (including the backbone network, etc.). "FLOPs" refer to the floating point operations of the method with an input of size [1 × 3 × 224 × 224]. All the numbers are obtained on a single NVIDIA RTX Titan GPU.

As seen, it is observed that our EMIDI achieves the best mIoU on three datasets, while requiring tolerable Params and FLOPs. Note that the transformer-based methods usually require more parameters and computation complexity. Even under such circumstance, our EMIDI only requires 1/2 and 2/3 of the FLOPs of RSSFormer and Segmenter, while the Params are 36.96 and 112.3 G Params less than these two existing transformer-based methods, respectively. These advantages can be attributed to our specially designed CST block and well prove the efficiency of the proposed method.

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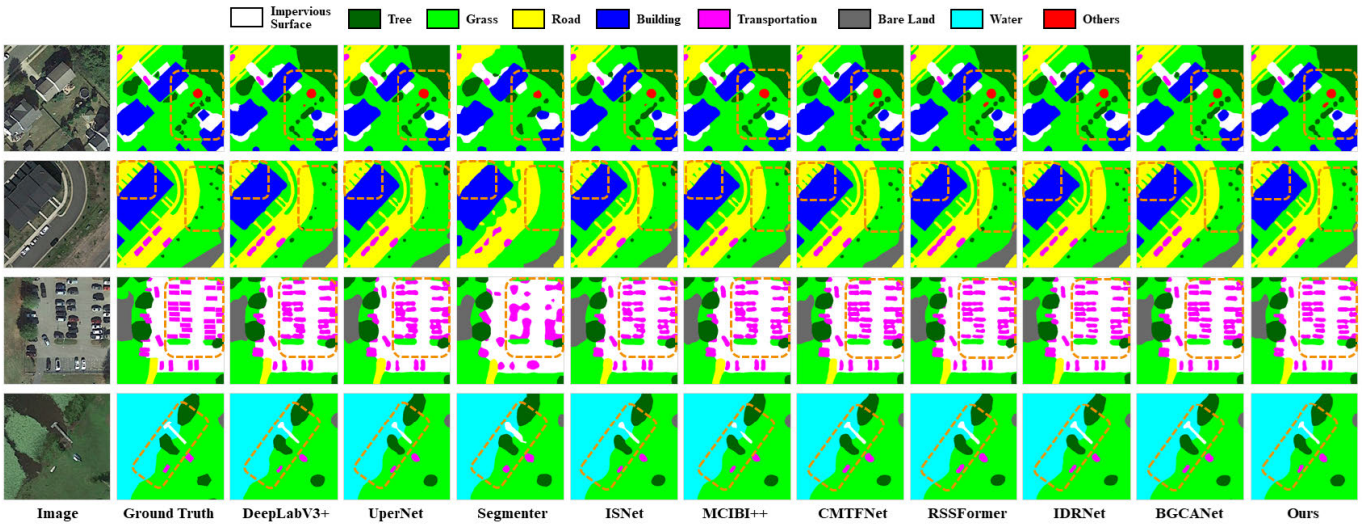


Fig. 9. Visualized comparison on GaoFen-2 dataset. (Left to Right) Input image, the ground-truth, the results from the state-of-the-art methods and our methods.

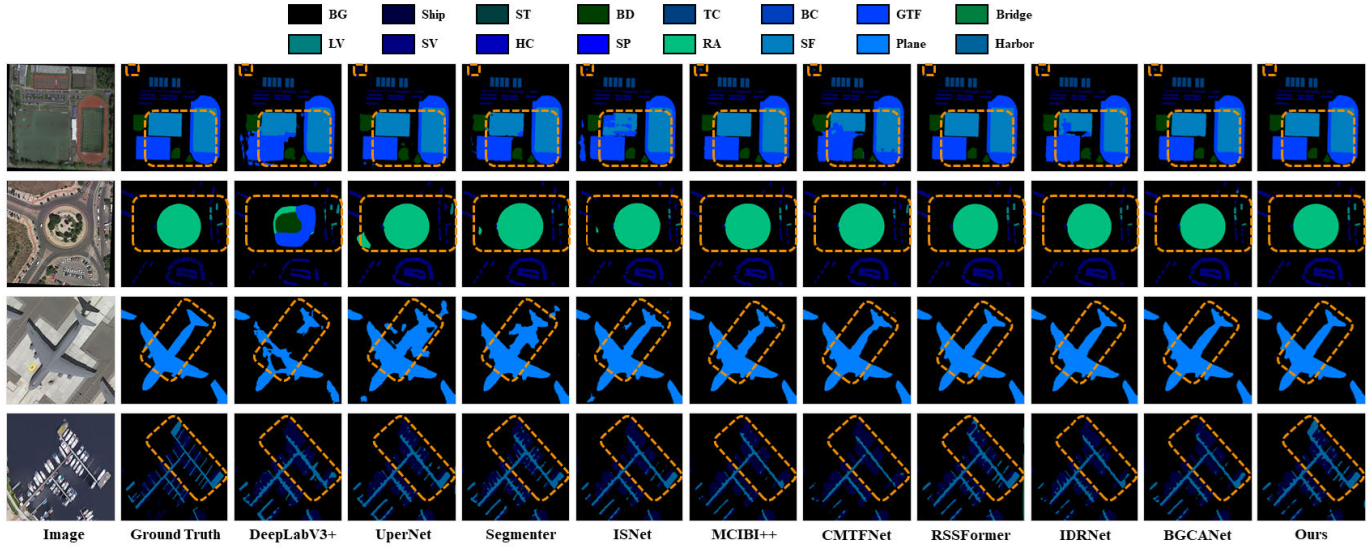


Fig. 10. Visualized comparison on iSAID dataset. (Left to Right) Input image, the ground-truth, the results from the state-of-the-art methods and our methods.

V. Conclusion

In this article, a semantic segmentation framework for encouraging the mutual interaction between dataset-level and image-level contextual cues is proposed for HRS images. Our framework comprises four key components: dataset-level context aggregation, DLFM, MLCI, and PFI-Net based on CST. To begin, we extract deep semantic features using a deep residual encoder and update the memory module to aggregate dataset-level contextual cues. Subsequently, the DLFM and MLCI modules are utilized for feature modulation and MLCI. Lastly, we feed the resulting contextual cues into PFI-Net for fine-grained fusion interaction.

Comparative experiments were conducted on three publicly available benchmarks of HRS images. The results show that EMIDI outperforms state-of-the-art semantic segmentation methods: U-Net, PSPNet, DeeplabV3+, DANet, Segmenter, OCRNet, ISNet, MCIBI++, CMTFNet, RSSFormer, IDRNet, BGCANet. Furthermore, ablation experiments and computational complexity analysis were performed on the three datasets. Extensive experiments demonstrate the effectiveness and robustness of our method. Moreover, our model exhibits certain advantages, such as its complex architecture. This can be improved by pruning the model, thereby enhancing inference speed and reducing the number of parameters.

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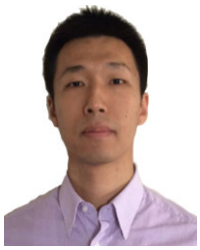
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